Separable Deep Graph Convolutional Network Integrated With CNN and Prototype Learning for Hyperspectral Image Classification

Yingjie Lu, *Graduate Student Member, IEEE*, Shaohui Mei, *Senior Member, IEEE*, Fulin Xu, Mingyang Ma, *Graduate Student Member, IEEE*, and Xiaofei Wang, *Graduate Student Member, IEEE*

***Abstract*— Graph convolutional networks (GCNs) have garnered extensive attention in the realm of hyperspectral image (HSI) classification. However, due to the problem of oversmoothing caused by deep GCN, most of the existing GCN-based methods are limited to constructing shallow networks, thus only able to extract superficial features. Moreover, when existing shallow GCNs extend to a more deeper structure, the number of learnable parameters increases linearly, thus leading to poor generalization performance under limited training samples. To address the aforementioned issues, a separable deep GCN integrated with convolutional neural network and prototype learning (SDGCP) is proposed for HSI classification, which can extract effective global structural information of HSI without increasing the number of trainable parameters. Specifically, the spectral and spatial features, adaptively selected by the attention module, are encoded into the structure of a graph by the graph encoder with the assistance of the pixel-to-region mapping obtained from the simple linear iterative clustering (SLIC). Then, a separable deep graph convolution module, composed of feature extraction and deep feature propagation, is adopted to capture the long-range contextual relationships from HSI encoded as graph data, which is combined with locally complementary information extracted by CNN after decoding. Finally, to further boost the performance of classification under limited labeled samples, prototype learning with regularization terms is utilized to enhance the intraclass compactness and interclass separability of feature representations. Extensive experiments on three standard HSI datasets demonstrate the superiority of the proposed SDGCP over the state-of-the-art (SOTA) methods.**

***Index Terms*— Attention mechanism, convolutional neural network (CNN), graph convolutional network (GCN), hyperspectral image (HSI) classification, prototype learning.**

I. Introduction

HYPERSPECTRAL imaging is a comprehensive Earth observation technology, which can obtain electromagnetic wave radiation information of ground objects. These spectrometer data captured by hyperspectral sensors typically provide abundant high-resolution spectral information, which is composed of hundreds of bands. Due to this characteristic, hyperspectral data are widely used in a variety of applications in the remote sensing field (e.g., precision agriculture [1], mineral detection [2], hydrological science [3], environmental science [4], and military applications [5]). Most of these applications primarily focus on the classification of each pixel in hyperspectral data.

Over the past few decades, hyperspectral image (HSI) classification has become one of the most important research fields in the hyperspectral community and has also attracted widespread attention [6], [7], [8], [9], [10], [11], [12], [13]. Intensive work has been performed to deal with HSI classification by utilizing machine learning methods over the last few decades. Neural networks [14], multinomial logistic regression [15], k-nearest neighbor [16], Bayesian estimation methods [17], and decision trees [18] have been investigated for this task. However, most of these methods heavily suffer from the Hughes phenomenon (i.e., the curse of dimensionality). To alleviate this issue, several dimension reduction (DR) methods, such as principal component analysis (PCA) [19], independent component analysis (ICA) [20], and linear discriminant analysis (LDA) [21], have been proposed to map the original HSI data into a lower dimensional space, which is easier to classify. Although these traditional machine learning methods probably acquire satisfactory performance, they are generally based on handcrafted spectral–spatial features, which heavily rely on professional knowledge and appropriate designs of manual features to a great extent.

Compared with conventional handcrafted methods, deep learning methods have a powerful representation capacity to extract discriminate and abstract features automatically via hierarchical architecture. Currently, the commonly used deep learning-based algorithms for HSI classification include stacked autoencoders (SAEs) [22], deep belief networks (DBNs) [23], convolutional neural networks (CNNs) [24], [25], [26], [27], [28], [29], recurrent neural networks (RNNs) [30], and generative adversarial networks (GANs) [31]. In particular, CNNs have shown prominent performance in classification accuracy and a great number of improvements have been made in CNN variants. Yang et al. [32] utilize two-channel deep CNN to extract spectral and spatial features contained in HSI. To make use of

Manuscript received 25 December 2023; revised 29 February 2024 and 9 April 2024; accepted 15 April 2024. Date of publication 17 April 2024; date of current version 1 May 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62171381 and Grant 62201445 and in part by the Project funded by China Postdoctoral Science Foundation under Grant 2022M722550. *(Corresponding author: Shaohui Mei.)*

Yingjie Lu, Shaohui Mei, Fulin Xu, and Xiaofei Wang are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: luyingjie@mail.nwpu.edu.cn; meish@nwpu.edu.cn; xfl@mail.nwpu.edu.cn; wangxiaofei2022@mail.nwpu.edu.cn).

Mingyang Ma is with the School of Information and Communications Engineering, Xi'an Jiaotong University, Xi'an, Shaanxi 710049, China (e-mail: mamingyang@mail.xjtu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3390575

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.

both spectral and spatial information derived from HSI data, 3-D CNNs have been applied for HSI classification [33]. However, as the depth of the network increases, the classification accuracy perhaps decreases. This decreasing-accuracy phenomenon can be moderated to some extent by introducing residual and dense connections [34], [35]. Inspired by human perception, numerous studies apply the attention mechanism to CNNs for HSI classification and achieve great progress [36]. These methods with smaller convolutional kernels face challenges in capturing long-range relationships, while some algorithms utilize large convolutional kernels to increase the receptive field of the CNN [37], [38], [39]. Moreover, an increasing number of methods have focused on cross-scene HSI classification [40], [41], [42], [43], [44].

Recently, graph convolutional networks (GCNs) have become an area of great interest due to the ability to perform convolution on graph structure data and have been applied to HSI classification. For instance, Qin et al. [45] proposed a semi-supervised GCN framework, which can perform message propagation between neighboring land cover targets based on spectral similarity and spatial information. Mou et al. [46] constructed a nonlocal graph by modeling relationships between different pixels in HSI and utilized GCN to propagate node information. However, these GCN-based methods utilize the entire HSI as input, resulting in significant computational costs. To address this issue, Wan et al. [47] proposed to substitute superpixel blocks for pixels to construct graphs, substantially reducing the number of graph nodes. In addition, Hong et al. [48] randomly sampled a specific number of pixels as graph nodes to construct mini-batch graphs, effectively mitigating computational burden. Although the abovementioned GCN-based methods have achieved significant performance in HSI classification, it is worth emphasizing that these methods are all based on shallow networks, which limits the model's ability to capture the global spatial information of HSI. Theoretically, deep networks can capture high-order global information through graph topology, but the oversmoothing problem limits the depth of GCN. This arises from the phenomenon that as the number of layers increases, the GCN can aggregate features of high-order neighbors, and the features of nodes will become closer when the threshold is exceeded, leading to a decrease in classification accuracy [49]. It is also notable that the input of these methods is a graph composed of superpixel nodes, which ignore local feature information. In addition, in such multiclass classification tasks, the softmax function is commonly used to generate probability distributions for each class. However, employing this approach for feature representation may result in poor discriminative ability between different categories, limiting improvements in classification performance [50].

To address the aforementioned challenges, a novel network named separable deep GCN integrated with CNN and prototype learning (SDGCP) framework is proposed for HSI classification, which includes a spectral–spatial attention subnetwork (S^2AsN), a global feature extraction subnetwork (GFEsN), a local feature extraction subnetwork (LFEsN), and a prototype learning module (PLM). Considering the rich spatial and spectral information in HSIs, a spatial–spectral attention mechanism is initially employed for data preprocessing. Subsequently, the hybrid network, comprising a separable deep GCN and CNN, extracts more discriminative complementary global and local features under limited labeled samples. After fusing these two features, prototype learning is implemented along with intraclass and interclass loss functions to enhance the discriminability of features, and abundant experiments on three datasets substantiate that SDGCP exhibits promising classification performance compared with other state-of-the-art (SOTA) CNN and GNN methods.

The primary contributions of this article are specified as follows.

1) In order to tackle the issue of oversmoothing caused by deep GCN, a separable deep graph convolution module (SDGCM) is proposed to decouple feature transformation from deep feature propagation, which not only effectively captures the global structural information of HSIs but also reduces the number of learnable parameters.
2) To enhance the robustness of deep spectral–spatial feature learning, an S^2AsN is proposed to perform the dimensionality reduction of raw HSI and adaptive selection of spectral and spatial features by emphasizing crucial information while suppressing unnecessary features.
3) To compensate for the deficiencies of SDGCM in pixel-level features, a simple LFEsN composed of CNN is employed to capture local features, which are then fused with the global features extracted by the proposed deep GCN.
4) To further improve the classification performance under limited labeled samples, a specialized loss function is designed for the PLM to enhance both interclass separability and intraclass compactness, which can compel the model to extract more discriminative features.

The rest part of this article is organized as follows. The structure of our proposed SDGCP is introduced in Section II. Section III details the classification results of different methods. Section IV exhibits the discussions. Conclusions are provided in Section V.

II. Proposed Method

A. Architecture of the Proposed SDGCP

The illustration of the proposed SDGCP is displayed in Fig. 1, which consists of an S^2AsN, a GFEsN, an LFEsN, and a PLM. First, for the original HSI, a simple linear iterative clustering (SLIC) is employed to achieve pixel-to-superpixel mapping, and dimensionality reduction and adaptive selection of spatial–spectral features are carried out by the S^2AsN. Second, the refined features are separately fed into GFEsN and LFEsN. GFEsN employs an SDGCM to extract global contextual features from the graph encoded as superpixels, while complementary local features are extracted through stacked layers of CNN in LFEsN. Third, to integrate the features extracted by the subnetworks of the two branches, a weighted sum is employed to fuse the output global and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.

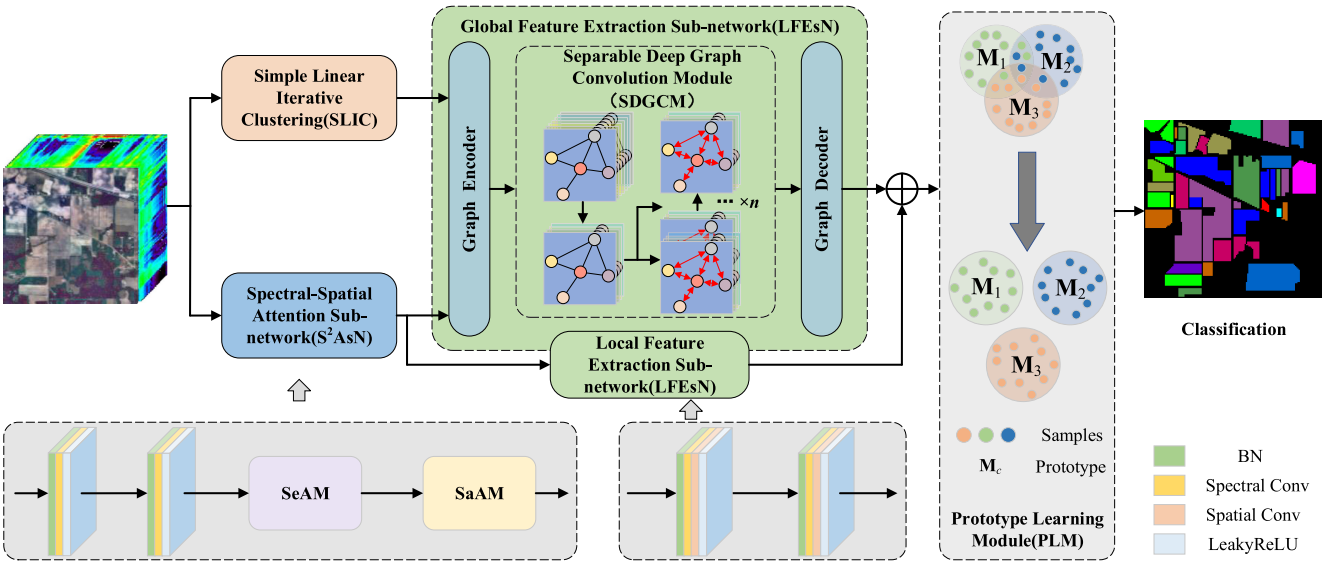


Fig. 1. Framework of our proposed SDGCP. It consists of the S^{2}AsN, GFEsN, LFEsN, and a PLM.

local features. Finally, to further enhance the discriminative capability of the features, PLM is utilized to accurately classify the integrated global and local features.

B. Spectral–Spatial Attention Subnetwork

Due to the presence of noise and redundant information in the original HSI, extracting features directly from it might lead to inaccurate results [51]. Therefore, S^{2}AsN initially preprocesses the original HSI and performs feature selection to enhance robust spectral features. The S^{2}AsN is described in detail as follows.

Given an input HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B denote the height, width, and number of bands of HSI, respectively, the input is first processed by two layers of spectral convolution for dimensionality reduction and denoising. The spectral convolutional layer is a specialized case of a standard convolutional layer with a kernel size of 1×1, which can be expressed by

$$\mathbf{X}^{l}_{h,w,m} = \phi\left(\sum_{b} \mathbf{K}^{l}_{i,j,b,m}\hat{\mathbf{X}}^{l-1}_{h,w,b} + \mathbf{B}^{l}_{i,j,b,m}\right) \tag{1}$$

where $\hat{\mathbf{X}}^{l-1}_{h,w,b} = \mathrm{BN}(\mathbf{X}^{l-1}_{h,w,b})$ is the normalized input at the lth layer; $\mathbf{X}^{l}_{h,w,m}$, $\mathbf{K}^{l}_{i,j,b,m}$, and $\mathbf{B}^{l}_{i,j,b,m}$ represent the output, kernel, and bias of the lth layer of spectral convolution, respectively; and ϕ denotes the LeakyReLU activation function. After processing by the spectral convolutional layer, the spectral information of different bands in HSI can be effectively connected, which will remove irrelevant spectral details and enhance the ability to learn spectral features.

Then, the attention mechanism [52] is introduced to construct a spectral–spatial attention module for refining the spatial and spectral features in HSI. The spectral attention module (SeAM) is depicted in Fig. 2. Assuming that the output of the spectral convolutional layer is represented as $\mathbf{X}^{\mathrm{o}} \in \mathbb{R}^{H\times W\times M}$, the SeAM aggregates spatial information of

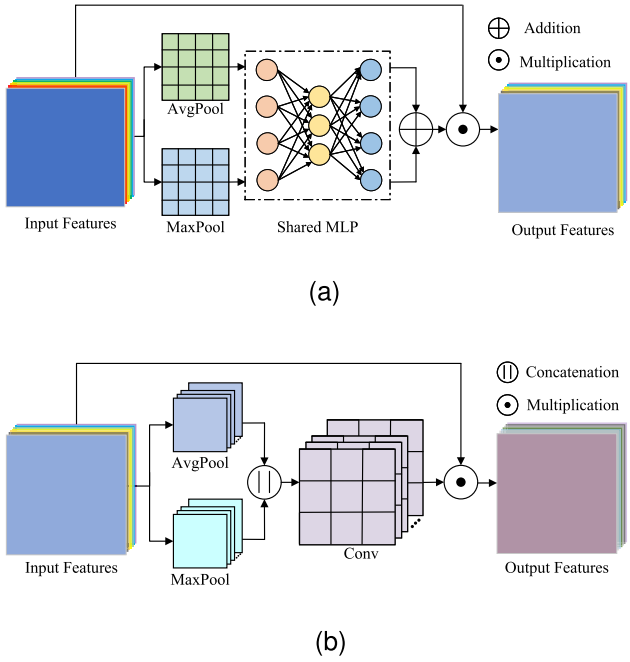


Fig. 2. Detailed structure display of (a) SeAM and (b) SaAM.

features from different perspectives using average pooling and max pooling to obtain $\mathbf{X}^{\mathrm{avg}}_{\mathrm{Spe}}$ and $\mathbf{X}^{\mathrm{max}}_{\mathrm{Spe}}$, where $\{\mathbf{X}^{\mathrm{avg}}_{\mathrm{Spe}}, \mathbf{X}^{\mathrm{max}}_{\mathrm{Spe}}\} \in \mathbb{R}^{1\times 1\times M}$. To fully capture the interaction between different channels, a shared multilayer perceptron (MLP) consisting of two fully connected layers with $\mathbf{W}_0 \in \mathbb{R}^{M\times(M/r)}$ and $\mathbf{W}_1 \in \mathbb{R}^{(M/r)\times M}$ is employed to generate two spectral attention maps, where r denotes the reduction ratio and is set to 8 in this context. After fusing these two attention maps, a spectral attention vector is obtained, which is element-wise multiplied with the input feature $\mathbf{X}^{\mathrm{o}}$ to yield the refined feature map, $\mathbf{X}^{\mathrm{o}}_{\mathrm{Spe}}$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.

In summary, the output of the SeAM can be represented as

$$\mathbf{X}_{\text{Spe}}^{\text{o}} = \mathbf{X} \odot \sigma\left(\delta\left(\mathbf{X}_{\text{Spe}}^{\text{avg}}\mathbf{W}_0\right)\mathbf{W}_1 + \delta\left(\mathbf{X}_{\text{Spe}}^{\text{max}}\mathbf{W}_0\right)\mathbf{W}_1\right) \quad (2)$$

where $\odot$, δ, and σ, respectively, represent the element-wise multiplication, the ReLU activation function, and the sigmoid activation function.

The structure of the spatial attention module (SaAM) is illustrated in Fig. 2. Unlike the SeAM, the SaAM connects the feature maps $X_{\text{Spa}}^{\text{avg}}$ and $X_{\text{Spa}}^{\text{max}}$ generated by average pooling and max pooling, respectively, where $\{\mathbf{X}_{\text{Spa}}^{\text{avg}}, \mathbf{X}_{\text{Spa}}^{\text{max}}\} \in \mathbb{R}^{H\times W\times 1}$. Then, the convolutional kernel is utilized to generate the spatial attention map, which is multiplied by the output $\mathbf{X}_{\text{Spe}}^{\text{o}}$ from the SeAM to obtain the refined feature map. Its formula can be written as

$$\mathbf{X}_{\text{Spa}}^{\text{o}} = \mathbf{X}_{\text{Spe}}^{\text{o}} \odot \sigma\left(\left[\mathbf{X}_{\text{avg}}, \mathbf{X}_{\text{max}}\right] * \mathbf{W}_3\right) \quad (3)$$

where $*$ is the convolution operation and $\mathbf{X}_{\text{Spa}}^{\text{o}}$ is the output of SaAM. By introducing the spectral–spatial attention mechanism, the spatial and spectral features in HSI can be adaptively selected, which suppresses unnecessary features simultaneously. The refined feature maps obtained through spectral and spatial attention demonstrate enhanced robustness to noisy input, thus facilitating subsequent feature extraction and classification tasks.

C. Global Feature Extraction Subnetwork

Existing GCN-based methods tend to construct networks that are shallow in depth, leading to feature extraction that remains confined to low-order neighboring node information. In addition, deeper networks can lead to the problem of oversmoothing, which in turn decreases classification accuracy. To address these issues, the SDGCM is proposed to construct a deep network for extracting global features from HSI while mitigating oversmoothing. In contrast to traditional deep GCNs, SDGCM separates feature extraction from feature propagation, significantly reducing the number of learnable parameters.

Before extracting the global information from HSI, it is crucial to encode it into a graph structure. However, directly considering each pixel of the entire HSI as a node in the graph will result in a large graph scale, which leads to high computational costs. Inspired by [51], superpixels segmented by SLIC are utilized as nodes for graph construction. To better align superpixels with homogeneous areas in HSI, we introduce a hyperparameter λ to adjust the size of superpixels. The relationship is given by $S = \lceil (H \times W)/\lambda \rceil$, where S represents the number of superpixels in the HSI and $\lceil \bullet \rceil$ denotes the ceiling function. The detailed process of constructing the graph is illustrated in Fig. 3. The pixel-to-superpixel mapping, which includes the mapping relation $\mathbf{Q} \in \mathbb{R}^{HW\times S}$ and the spatial structure information $\mathbf{M} \in \mathbb{R}^{S\times S}$, is obtained by segmenting the original HSI with SLIC. Given an HSI $\mathbf{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n]$, the expression for $\mathbf{Q}$ is as follows:

$$\mathbf{Q}_{i,j} = \begin{cases} 1, & \text{if } \boldsymbol{x}_i \in S_j \\ 0, & \text{if } \boldsymbol{x}_i \notin S_j \end{cases} \quad (4)$$

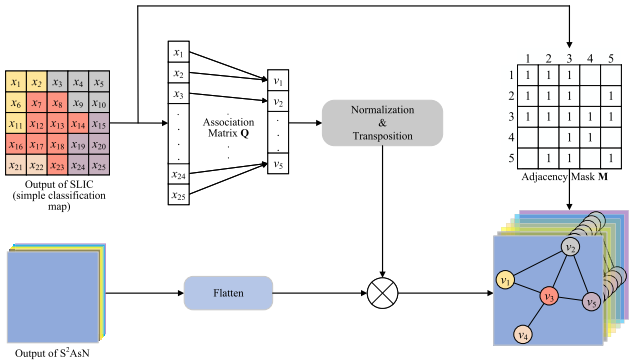


Fig. 3. Flowchart of the graph encoder. The original HSI is processed by SLIC to generate a simple classification map, from which both the association matrix $\mathbf{Q}$ between pixels and superpixels and the adjacency mask matrix $\mathbf{M}$ between the superpixels can be extracted. The association matrix $\mathbf{Q}$ is normalized and transposed to encode the output of S^2AsN into the superpixel node features such that the graph structure is constructed with the assistance of the mask matrix $\mathbf{M}$.

where $\boldsymbol{x}_i$ $(i = 1, 2, \ldots, n)$ and S_j $(j = 1, 2, \ldots, S)$ denote the ith pixel and jth superpixel, respectively. Based on the output of S^2AsN and the mapping relationship $\mathbf{Q}$, the node features of the superpixel graph can be expressed as follows:

$$\mathbf{V} = \text{Encoder}\left(\mathbf{X}_{\text{Spa}}^{\text{o}}; \mathbf{Q}\right) = \overline{\mathbf{Q}}^T \text{Flatten}\left(\mathbf{X}_{\text{Spa}}^{\text{o}}\right) \quad (5)$$

where $\overline{\mathbf{Q}}$ denotes the normalized $\mathbf{Q}$ by column and $\mathbf{V} = [\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_S]$ is the node features of superpixels. In contrast to the graph encoding module, the graph decoding module transforms graph data back into 3-D cube data, which is used for subsequent feature fusion and classification tasks. The specific formula can be represented as

$$\tilde{\mathbf{X}} = \text{Decoder}(\mathbf{V}; \mathbf{Q}) = \text{Reshape}(\mathbf{QV}) \quad (6)$$

where $\tilde{\mathbf{X}}$ denotes the 3-D cube data converted from node features. In addition to constructing node features for the graph, building a complete graph also encompasses neighbor node selection and the establishment of an adjacency matrix. In this article, spatially adjacent superpixels are chosen as neighboring nodes, and the adjacency matrix of the graph is represented by the radial basis function

$$\mathbf{A}_{ij} = \begin{cases} \exp\left(-\gamma \left\| \boldsymbol{v}_i - \boldsymbol{v}_j \right\|^2\right), & \text{if } \mathbf{M}_{ij} = 1 \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

where $\mathbf{M}_{ij}$ represents the connection between node i and node j, with a value of 1 indicating a connection and 0 indicating the opposite. The parameter γ controls the width of the radial basis function, and here, it is set to 0.01.

After encoding 3-D grid data into graph data, the SDGCM is employed to extract global contextual information from the HSI. Before introducing SDGCM, the GCN layer is discussed. Considering an undirected graph $\mathbf{X}_P$ with its adjacency matrix $\mathbf{A}_P$, the propagation equation for the lth GCN layer is as follows:

$$\mathbf{X}_P^{(l+1)} = \text{ReLU}\left(\hat{\mathbf{A}}_P \mathbf{X}_P^{(l)} \mathbf{\Theta}^{(l)}\right) \quad (8)$$

$$\hat{\mathbf{A}}_P = \widetilde{\mathbf{D}}_P^{-1/2} \widetilde{\mathbf{A}}_P \widetilde{\mathbf{D}}_P^{-1/2} \quad (9)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.

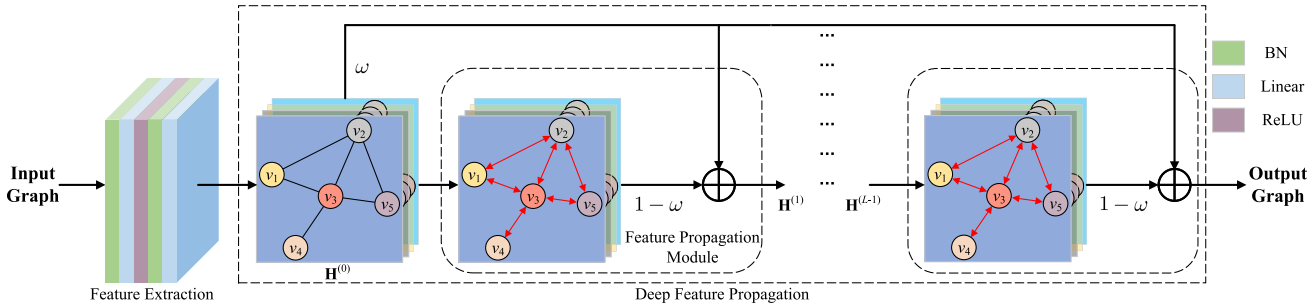


Fig. 4. Flowchart of SDGCM, which comprises feature extraction and deep feature propagation operations. The input is the superpixel graph, $\mathbf{H}^{(0)}$ represents the graph after feature extraction, and $\mathbf{H}^{(l)}$ signifies the node features after the lth feature propagation module. ω is a hyperparameter utilized to adjust the proportion of initial graph features in the residual connection.

where $\mathbf{X}_P^{(l+1)}$ and $\mathbf{X}_P^{(l)}$ denote the output and input of layer l, respectively, and $\widetilde{\mathbf{D}}_P$ is a diagonal matrix of $\widetilde{\mathbf{A}}_P$ with self-loops, representing the number of edges connected to the nodes. $\mathbf{\Theta}^{(l)}$ is learnable parameter matrix of the lth layer. As the number of layers increases, this smooth feature propagation makes it challenging to distinguish features between nodes of different classes. To alleviate the oversmoothing problem, the residual connection is introduced to construct deep GCN. However, merely increasing the network depth would result in an increase in the number of learnable parameters, but sacrificing computational time cost to achieve higher accuracy is not our original intention. Therefore, feature extraction is decoupled from feature propagation to construct the SDGCM. This module first performs feature extraction on the superpixel graph and then conducts rapid feature propagation, which not only alleviates the issue of oversmoothing but also reduces the number of learnable parameters in deep GCN. The framework of SDGCM is illustrated in Fig. 4. Feature extraction is initially applied to each node of the input superpixel graph, and its formula is as follows:

$$\mathbf{V}^{(1)} = \mathrm{ReLU}\left(\tilde{\mathbf{V}}\mathbf{W}^{(0)} + \mathbf{B}^{(0)}\right) \tag{10}$$

$$\mathbf{H}^{(0)} = \tilde{\mathbf{V}}^{(1)}\mathbf{W}^{(1)} + \mathbf{B}^{(1)} \tag{11}$$

where $\tilde{\mathbf{V}}$ and $\tilde{\mathbf{V}}^{(1)}$ represent the normalized input features for $\mathbf{V}$ and $\mathbf{V}^{(1)}$, respectively. $\mathbf{W}^{(0)} \in \mathbb{R}^{M\times M_h}$ and $\mathbf{W}^{(1)} \in \mathbb{R}^{M_h\times M_o}$ are learnable parameter matrices, where M_h and M_o are the number of hidden and output channels, respectively. $\mathbf{B}^{(0)}$ and $\mathbf{B}^{(1)}$ denote bias matrices. It is evident that the proposed SDGCM has a parameter count comparable to that of only two layers of GCN. After performing the feature extraction operation, multiple feature propagation modules are utilized to facilitate long-range information interaction among graph nodes. The mathematical formula for the lth feature propagation module is defined as

$$\mathbf{H}^{(l)} = (1-\omega)\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}\mathbf{H}^{(l-1)} + \omega\mathbf{H}^{(0)}. \tag{12}$$

In this module, we introduce an initial residual connection that retains a certain amount of original information during each feature propagation, which alleviates the oversmoothing issue caused by deep GCN. By stacking multiple feature propagation modules, the central node can effectively aggregate features from higher order neighboring nodes, thereby capturing the global spatial context information in the HSI. From SDGCM, it is evident that the only learnable parameters are $\mathbf{W}^{(0)}$ and $\mathbf{W}^{(1)}$. In contrast to deep GCN, no additional parameters are introduced at each layer during the deep feature propagation process. This design enables the GFEsN to rapidly extract global information.

D. Local Feature Extraction Subnetwork

Considering that superpixel nodes overlook the relationships among their internal pixels, CNN is integrated into the network to extract local and spectral features from HSI. These features are then fused with the global features extracted by GFEsN to enhance feature discriminability and the robustness of the network. Due to the large number of parameters in standard 2-D convolution, it may result in overfitting issues under limited labeled samples. Therefore, LFEsN utilizes two layers of lightweight spectral convolution and spatial convolution to reduce the parameters and enhance the robustness. Spectral convolution, essentially a 1-D convolution, is used to capture relationships among HSI channels, and its computational formula represents a spectral convolutional layer without batch normalization and activation functions. Spatial convolution, on the other hand, conducts individual convolution operations on each input channel, with each channel having its dedicated convolutional kernel. Its mathematical expression is depicted as

$$\mathbf{X}^*_{h,w,m} = \sum_{i,j} \mathbf{K}_{i,j,m}\mathbf{X}_{h,w,m} + \mathbf{B}_{i,j,m} \tag{13}$$

where $\mathbf{X}_{h,w,m}$ is the input of the spatial convolution and $\mathbf{X}^*_{h,w,m}$, $\mathbf{K}_{i,j,m}$, and $\mathbf{B}_{i,j,m}$ represent the output, kernel, and bias of the spatial convolution, respectively. By employing spectral–spatial convolution, LFEsN can extract local information in HSI with fewer parameters, which complements the global information extracted by GFEsN.

E. Prototype Learning Module

The input to the PLM is the fusion of the output features from two feature extraction subnetworks, and the fusion formula is represented as

$$\mathbf{H}^* = (1-\alpha)\mathbf{H}_\mathrm{G} + \alpha\mathbf{H}_\mathrm{L} \tag{14}$$

where α is a hyperparameter that controls the fusion ratio of the two types of features and $\mathbf{H}_\mathrm{G}$ and $\mathbf{H}_\mathrm{L}$ represent the global

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.

features extracted by GFEsN and the local features extracted by LFEsN, respectively.

Specifically, let C and N denote the number of classes and the number of samples, respectively. The prototype representations of the samples are denoted as $\mathbf{P} = \{\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_C\}$, where $\mathbf{p}_c$ represents the prototype of the cth class. For a sample $\mathbf{x}_i$, its probability of belonging to the cth class is expressed as

$$p(\mathbf{x}_i \in \mathbf{p}_c|\mathbf{x}_i) = \frac{e^{-\theta d(f(\mathbf{x}_i),\mathbf{p}_c)}}{\sum_{k=1}^{C} e^{-\theta d(f(\mathbf{x}_i),\mathbf{p}_k)}} \tag{15}$$

where $i \in \{1, 2, \ldots, N\}$, $c \in \{1, 2, \ldots, C\}$, $f(\mathbf{x}_i)$ represents the node embedding obtained after feature extraction, $d(f(\mathbf{x}_i), \mathbf{p}_c)$ denotes the distance between the node embedding and the cth prototype, and θ is a hyperparameter. Based on the probability $p(\mathbf{x}_i \in \mathbf{p}_c|\mathbf{x}_i)$, the distance-based cross-entropy (DCE) loss function can be derived as

$$L_{\mathrm{DCE}}(\mathbf{Y}, \mathbf{P}) = -\frac{1}{N \times C} \sum_{i=1}^{N} \sum_{c=1}^{C} y_{i,c} \log p(\mathbf{x}_i \in \mathbf{p}_c|\mathbf{x}_i) \tag{16}$$

where $\mathbf{Y}$ denotes the ground truth and $y_{i,c}$ denotes the cth element of the ith label in $\mathbf{Y}$.

During the testing phase, for any test sample $\hat{\mathbf{x}}$, its predicted label is determined by the distances between the extracted features and each prototype representation. The specific formula is expressed as

$$\hat{\mathbf{y}} = \arg\min_{j} \left(d\left(f\left(\hat{\mathbf{x}}\right), \mathbf{p}_j\right)\right) \tag{17}$$

where $\hat{\mathbf{y}}$ is the predicted label for $\hat{\mathbf{x}}$.

To further enhance the discriminative power of features under limited labeled samples, prototype learning is introduced with two designed regularizers: a focusing regularizer and a separating regularizer. The focusing regularizer loss minimizes the distance between the features of the samples and their prototypes, thereby making the distribution of features from the same class more concentrated. It can be represented by the following formula:

$$L_{\mathrm{foc}} = \frac{1}{N} \sum_{c=1}^{C} \sum_{j=1}^{N_c} \left\| x_{cj} - p_c \right\|_2^2 \tag{18}$$

where $N = \sum_{c=1}^{C} N_c$, in which N_c denotes the number of samples in the cth class. In addition, to increase the interclass distance, the separating regularizer is used to increase the distance between samples and other prototypes. It is formulated as

$$L_{\mathrm{sep}} = \frac{N \times (C-1)}{\sum_{c \neq k} \sum_{j=1}^{N_c} \left\| x_{kj} - p_c \right\|_2^2}. \tag{19}$$

With these two regularizer losses, we can optimize the interclass and intraclass distances of samples, enhancing both intraclass compactness and interclass separability to further improve the classification performance. The total loss function is as follows:

$$L_{\mathrm{total}} = L_{\mathrm{DCE}} + Q_1 \cdot L_{\mathrm{foc}} + Q_2 \cdot L_{\mathrm{sep}} \tag{20}$$

where Q_1 and Q_2 are hyperparameters for the intensity of regularizations, L_{foc} and L_{sep}, respectively.

TABLE I
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE INDIAN PINES DATASET

No.	Class	Train	Val	Test	Total
1	Alfalfa	1	1	44	46
2	Corn-notill	15	15	1398	1428
3	Corn-mintill	9	9	812	830
4	Corn	3	3	231	237
5	Grass-pasture	5	5	473	483
6	Grass-trees	8	8	714	730
7	Grass-pasture-mowed	1	1	26	28
8	Hay-windrowed	5	5	468	478
9	Oats	1	1	18	20
10	Soybean-notill	10	10	952	972
11	Soybean-mintill	25	25	2405	2455
12	Soybean-clean	6	6	581	593
13	Wheat	3	3	199	205
14	Woods	13	13	1239	1265
15	Buildings-Grass-Trees	4	4	378	386
16	Stone-Steel-Towers	1	1	91	93
	Total	110	110	10029	10249

III. EXPERIMENTS

To evaluate the classification performance of the proposed SDGCP, three well-known HSI datasets, i.e. Indian Pines, Salinas, and WHU-Hi-LongKou, are adopted to conduct the experiments. The training ratios for these three datasets are set to 1%, 0.2%, and 0.1% in a randomized division, and the division results are consistent for all methods. Furthermore, three evaluation metrics, namely, overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ), are employed to assess the classification results.

A. Datasets

1) Indian Pines Dataset: The Indian Pines dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in 1992. It contains 220 bands with the size of 145 × 145 pixels. After removing 20 noisy bands, the remaining 200 bands and 16 labeled classes of land covers are used for classification. Table I shows the number of training sets, validation sets, and test sets in different classes.

2) Salinas Dataset: The Salinas dataset was captured over the Salinas Valley, which contains 224 bands and 512 × 217 pixels with a spatial resolution of 3.7 × 3.7 m. After removing noisy bands, 204 bands were reserved, and 54 129 samples were labeled into 16 classes. Table II presents each category and dataset division.

3) WHU-Hi-LongKou Dataset: The WHU-Hi-LongKou dataset was captured by Headwall Nano-Hyperspec imaging sensor in Longkou Town, Hubei province, China, which contains 270 bands from 400 to 1000 nm, and the size of this HSI is 550 × 400 pixels with nine different categories. Table III shows the training, validation, and testing samples.

B. Experimental Settings

Nine different SOTA deep learning methods for HSI classification are employed for comparison with SDGCP, including multiscale 3-D deep CNN (M3D-DCNN) [33], deep feature fusion network (DFFN) [35], residual spectral–spatial attention

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.

TABLE II
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE SALINAS DATASET

No.	Class	Train	Val	Test	Total
1	Brocoli_green_weeds_1	5	5	1999	2009
2	Brocoli_green_weeds_2	8	8	3710	3726
3	Fallow	4	4	1968	1976
4	Fallow_rough_plow	3	3	1388	1394
5	Fallow_smooth	6	6	2666	2678
6	Stubble	8	8	3943	3959
7	Celery	8	8	3563	3579
8	Grapes_untrained	23	23	11225	11271
9	Soil_vinyard_develop	13	13	6177	6203
10	Corn_senesced_green_weeds	7	7	3264	3278
11	Lettuce_romaine_4wk	3	3	1062	1068
12	Lettuce_romaine_5wk	4	4	1919	1927
13	Lettuce_romaine_6wk	2	2	912	916
14	Lettuce_romaine_7wk	3	3	1064	1070
15	Vinyard_untrained	15	15	7238	7268
16	Vinyard_vertical_trellis	4	4	1799	1807
	Total	116	116	53897	54129

TABLE III
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE WHU-HI-LONGKOU DATASET

No.	Class	Train	Val	Test	Total
1	Corn	35	35	34441	34511
2	Cotton	9	9	8356	8374
3	Sesame	4	4	3023	3031
4	Broad-leaf soybean	64	64	63084	63212
5	Narrow-leaf soybean	5	5	4141	4151
6	Rice	12	12	11830	11854
7	Water	68	68	66920	67056
8	Roads and houses	8	8	7108	7124
9	Mixed weed	6	6	5217	5229
	Total	211	211	204120	204542

network (RSSAN) [36], GCN [53], multiscale dynamic GCN (MDGCN) [47], context-aware dynamic GCN (CADGCN) [54], CNN-enhanced GCN (CEGCN) [51], deep attention GCN (DAGCN) [55], and deep network with adaptive graph structure integration (DNAGSI) [56]. To ensure a fair comparison, all GCN-based methods employ superpixel graphs as input, with segmentation scales λ of 50, 300, and 300 for the Indian Pines, Salinas, and WHU-Hi-LongKou datasets, respectively. The remaining methods maintain the same hyperparameter settings as in the original papers. All the codes are implemented in PyTorch on RTX3090 (24 GB) GPUs.

For SDGCP, the number of kernels in the spatial convolution of S^2AsN is set to 128. In GFEsN, the weight matrices $\mathbf{W}^{(0)}$ and $\mathbf{W}^{(1)}$ used in feature extraction have shapes of 128×64 and 64×64, respectively. The number of layers for feature propagation is set to 64, with an initial residual connection coefficient ω set to 0.1. In LFEsN, two layers of spectral–spatial convolution are employed, with 128 and 64 kernels for spectral convolution and a 5×5 kernel size for spatial convolution. The hyperparameters Q_1 and Q_2 controlling the intensity of focusing regularizer and separating regularizer in the PLM are set to 0.001 and 0.1, respectively. The network is trained using the Adam optimizer with a learning rate of 0.001, and the number of training iterations is set to 300.

C. Classification Performance

Tables IV–VI show the classification results of the different models on each dataset, including OA, AA, κ, and classification accuracy for each category. All classification results are composed of the AA of ten experiments and their standard deviations, and the highest values of different models under the same standard are shown in bold. The classification maps obtained by these methods are illustrated in Figs. 5–7, with different colors indicating different ground objective categories. The highlighted white boxes showcase key regions and their corresponding enlarged images.

The first experiment is performed on the Indian Pines dataset and the classification results are shown in Table IV. Due to the limited number of training samples and the extreme imbalance of samples in the Indian Pines dataset, the performance of comparative algorithms may not be satisfactory, but the proposed SDGCP achieves an AA of 92.32%, and both OA and κ surpass those of other algorithms. It can be observed that the classification accuracy of CNN-based methods is relatively low. This is because CNNs cannot capture the features of the samples adequately when the training samples are limited. In contrast, GCN-based methods can achieve higher classification results under the same training ratio, in which MDGCN utilizes multiscale features and DAGCN and DNAGSI capture spatial context information in HSI through deep GCN. However, from the classification map in Fig. 5, it is evident that GCN-based methods exhibit poor performance in classifying the region boundaries, which is attributed to the limitations of the superpixel segmentation techniques employed. Affected by this factor, the proposed SDGCP may exhibit misclassifications when dealing with certain small targets. Overall, by appropriately combining the local features extracted by LFEsN with the global features extracted by GFEsN, SDGCP can preserve intricate details at the boundaries while achieving a smooth classification map and higher classification accuracy.

For the Salinas dataset, the proposed SDGCP model exhibits remarkable classification performance, outperforming the other comparative algorithms, with OA, AA, and κ all exceeding 99%, and the results are shown in Table V. DFFN, which integrates features from multiple layers, achieves superior classification results compared to the other two CNN methods. CADGCN has lower classification accuracy than GCN, due to the mismatch between the constructed dynamic graph and the Salinas dataset. The OA of MDGCN, DAGCN, and DNAGSI is lower than that of CEGCN, which is attributed to the fact that using only superpixels as nodes of the graph will lose detailed features at region boundaries. SDGCP integrates pixel-level detailed features with complementary superpixel-level contextual features and enhances feature discriminability through PLM, which contributes to superior classification performance. The classification maps obtained by these methods are illustrated in Fig. 6. It is evident that SDGCP significantly reduces misclassification.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE INDIAN PINES DATASET

Class	M3D-DCNN	DFFN	RSSAN	GCN	MDGCN	CADGCN	CEGCN	DAGCN	DNAGSI	SDGCP
1	0.00±0.00	20.45±11.77	5.00±7.10	93.93±1.05	81.19±21.99	75.72±29.47	13.88±9.20	94.61±1.48	**96.85±2.70**	95.04±2.44
2	43.89±6.66	69.59±4.40	60.50±22.61	80.28±7.30	79.80±9.74	75.18±8.25	82.48±8.22	79.91±12.62	82.34±8.70	**88.65±5.55**
3	14.41±8.17	57.92±10.80	31.13±19.76	77.71±8.47	79.16±9.47	72.91±11.32	71.30±14.42	79.82±6.49	74.77±10.65	**83.73±7.21**
4	6.61±7.53	61.41±12.23	14.01±13.20	87.81±9.03	91.09±9.97	57.20±21.53	63.70±19.56	83.93±16.62	**93.39±7.27**	93.13±6.66
5	23.92±7.73	73.92±7.41	71.12±10.19	75.55±11.58	81.85±8.88	73.28±15.51	**82.93±10.15**	80.79±9.92	80.03±7.79	82.63±8.13
6	86.90±2.84	92.13±4.38	92.44±1.97	91.01±5.06	94.13±4.04	87.12±6.66	97.39±1.21	90.95±4.61	93.89±4.50	**97.65±1.98**
7	0.00±0.00	47.69±25.70	1.54±1.88	90.00±30.00	90.00±30.00	99.63±1.11	80.00±25.94	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
8	97.48±1.34	89.83±5.84	77.30±38.71	98.00±2.66	**99.98±0.06**	79.41±15.90	98.87±2.56	98.13±2.72	94.81±9.92	**99.98±0.06**
9	0.00±0.00	0.00±0.00	0.00±0.00	67.87±35.48	80.50±20.10	75.23±26.92	55.61±27.6	79.44±21.99	78.39±35.09	**87.37±15.29**
10	44.09±7.96	70.41±7.18	49.44±25.16	80.21±7.82	81.04±6.12	77.63±9.56	82.08±8.26	79.82±7.32	83.70±4.26	**85.83±6.31**
11	82.83±3.30	77.51±5.23	84.00±7.63	87.02±5.65	92.68±6.43	82.76±5.25	91.35±5.11	91.20±6.18	92.52±5.81	**94.96±5.87**
12	11.04±11.24	43.48±11.33	33.22±23.34	70.71±9.08	82.15±9.28	60.71±18.3	62.90±11.88	81.79±9.30	78.39±10.65	**88.68±10.21**
13	58.38±28.41	96.55±2.43	56.04±32.62	96.86±1.91	97.16±1.69	96.96±2.07	98.15±1.79	95.61±3.91	97.75±1.74	**99.10±0.97**
14	91.97±2.73	88.21±4.34	94.26±2.68	97.05±4.27	96.34±4.44	94.88±4.72	**99.93±0.12**	96.76±3.81	94.82±6.23	97.74±3.23
15	17.20±5.47	57.68±5.81	37.74±19.46	77.10±10.77	86.57±9.15	69.34±17.97	72.58±18.63	86.00±11.02	85.15±11.82	**88.59±11.72**
16	85.17±5.09	94.61±3.21	68.54±36.78	87.83±20.65	68.00±19.82	82.64±12.21	71.60±27.60	88.49±20.52	92.68±6.86	**93.99±7.72**
OA(%)	57.90±2.37	73.87±2.32	66.37±6.55	85.08 ±2.60	87.98±1.82	79.54±1.56	85.58±2.52	87.30±2.50	87.77±2.43	**91.92±1.88**
AA(%)	41.49±2.74	65.09±1.69	48.52±9.51	84.93 ±3.28	86.35±2.56	78.79±1.59	76.55±4.50	87.95±1.70	88.72±2.55	**92.32±1.38**
$\kappa \times 100$	50.58±2.97	70.22±2.62	60.74±8.25	83.03±2.91	86.29±2.04	76.70±1.71	83.49±2.88	85.53±2.83	86.07±2.75	**90.79±2.13**

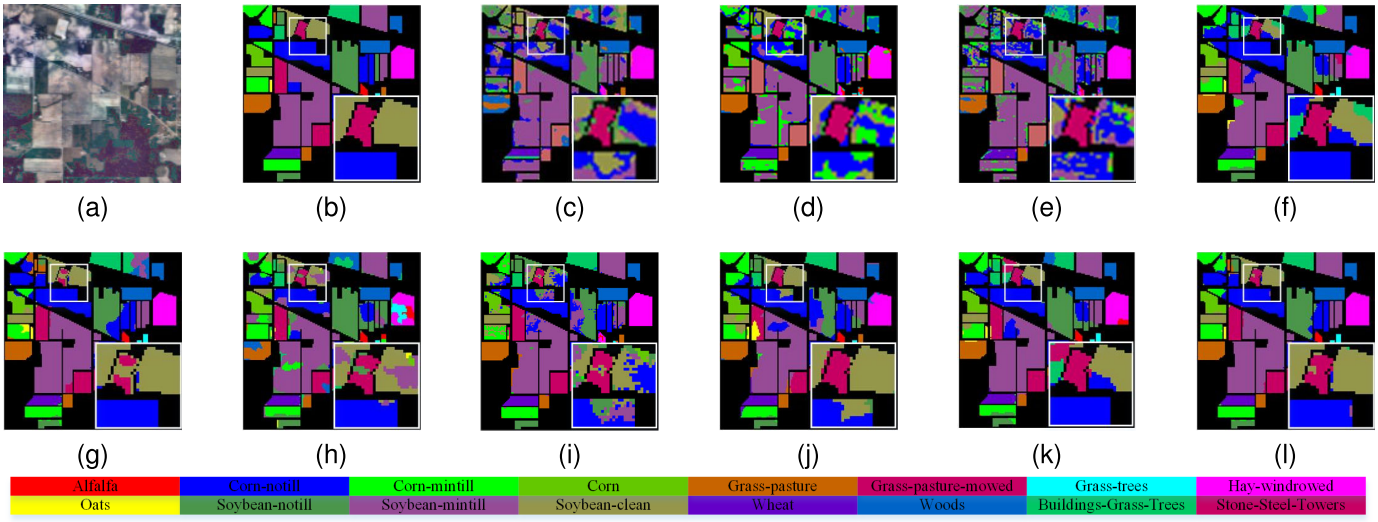


Fig. 5. False-color image, ground truth, and classification maps obtained by different methods on the Indian Pines dataset. (a) False-color image. (b) Ground Truth. (c) M3D-DCNN. (d) DFFN. (e) RSSAN. (f) GCN. (g) MDGCN. (h) CADGCN. (i) CEGCN. (j) DAGCN. (k) DNAGSI. (l) SDGCP.

As shown in Table VI, the accuracies obtained by SDGCP surpass all the compared methods. When compared to the results obtained from the other two datasets, M3D-DCNN, DFFN, and RSSAN showcase comparable classification accuracy on the WHU-Hi-LongKou dataset, all around 91%. This also indicates that DFFN exhibits stronger robustness when dealing with a smaller number of training samples compared to the other two CNN-based methods. CADGCN shows improved results compared to GCN, possibly due to the availability of more training samples in this dataset. Apart from the proposed SDGCP, CEGCN achieves the highest classification accuracy among GCN-based methods. This can be attributed to the increased number of training samples, which enhances the effectiveness of the local features extracted by the CNN. Although the proposed SDGCP achieves higher classification accuracy compared to other comparative algorithms, there are still certain classes (such as classes 5 and 9) with lower classification accuracy compared to other GCN-based methods. This indicates that while SDGCP can capture high-order neighbor information for nodes, it may not be universally effective for all land cover categories. From the classification result map in Fig. 7, it can be seen that the proposed SDGCP is closer to the ground truth.

IV. DISCUSSION

A. Ablation Study

To validate the effectiveness of different modules within the proposed SDGCP, a series of ablation experiments is conducted on three datasets. Specifically, a two-layer GCN combined with softmax-based cross-entropy loss is established as a baseline. On this basis, the typical GCN is replaced by the proposed GFEsN, and PLM is employed as an alternative to the softmax cross-entropy loss for pixel-wise classification of HSI. Subsequently, S^2AsN and LFEsN are introduced into the network to further assess their impact on the classification performance. The experimental results are illustrated in Table VII. All the results represent the mean of ten repeated experiments,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.

TABLE V
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE SALINAS DATASET

Class	M3D-DCNN	DFFN	RSSAN	GCN	MDGCN	CADGCN	CEGCN	DAGCN	DNAGSI	SDGCP
1	53.56 ± 31.16	94.00 ± 3.31	87.64 ± 11.65	95.61±5.39	**100.00±0.00**	99.99±0.02	99.87±0.22	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
2	98.37 ± 1.02	92.29 ± 10.78	95.69 ± 2.58	**100.00±0.00**	**100.00±0.00**	99.98±0.06	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
3	31.68 ± 12.29	72.07 ± 12.78	49.33 ± 26.26	**100.00±0.00**	**100.00±0.00**	95.02±10.73	99.68±0.72	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
4	98.24 ± 0.88	94.88 ± 4.60	97.13 ± 1.84	89.32±8.61	89.86±11.11	**99.62±0.45**	99.15±0.65	87.97±13.07	85.63±13.61	98.89±1.68
5	95.89 ± 1.25	86.27 ± 7.00	73.91 ± 35.28	96.29±5.37	94.24±10.71	97.49±0.46	97.45±5.53	94.45±10.94	94.51±10.79	**97.64±4.49**
6	99.78 ± 0.23	92.72 ± 5.97	98.64 ± 1.54	98.50±2.18	99.74±0.35	99.66±0.47	**99.96±0.04**	99.75±0.35	99.91±0.07	99.93±0.05
7	99.69 ± 0.26	90.19 ± 7.03	96.37 ± 2.11	99.24±0.71	**99.93±0.07**	98.59±0.98	99.92±0.08	**99.93±0.09**	99.91±0.07	**99.93±0.07**
8	84.98 ± 5.42	71.91 ± 3.29	76.65 ± 14.01	99.81±0.43	99.66±0.75	92.83±3.38	96.79±2.17	99.80±0.39	99.12±1.55	**99.82±0.49**
9	99.97 ± 0.03	95.02 ± 1.37	79.23 ± 39.62	99.98±0.05	**100.00±0.00**	99.81±0.34	99.99±0.03	**100.00±0.00**	99.99±0.03	**100.00±0.00**
10	39.01 ± 19.11	76.34 ± 5.40	51.46 ± 41.92	88.38±4.67	92.99±4.11	89.69±5.55	93.56±1.98	93.96±2.42	94.71±3.47	**96.57±1.97**
11	0.00 ± 0.00	77.12 ± 14.98	4.19 ± 8.38	94.94±6.78	**99.80±0.33**	89.87±5.17	99.33±0.46	99.47±1.00	96.85±5.63	99.76±0.38
12	87.61 ± 11.14	92.78 ± 7.99	55.78 ± 45.56	88.02±6.55	98.32±4.41	99.68±0.38	**99.97±0.05**	98.39±4.43	91.16±19.34	99.95±0.13
13	0.00 ± 0.00	77.37 ± 11.36	19.43 ± 38.86	83.92±15.83	86.71±16.01	93.23±9.43	**99.65±0.39**	79.49±13.81	83.18±19.97	98.24±3.06
14	7.94 ± 12.09	93.17 ± 4.50	17.04 ± 34.07	92.89±9.89	88.47±12.22	91.08±7.07	98.21±1.38	88.01±11.78	90.06±9.54	**98.74±1.03**
15	36.17 ± 13.38	72.92 ± 10.07	62.95 ± 32.98	97.80±3.31	98.89±1.37	91.90±6.48	97.06±2.69	98.88±1.42	98.81±1.56	**99.58±0.28**
16	37.94 ± 18.65	57.62 ± 10.14	79.06 ± 12.88	97.51±7.47	99.81±0.37	89.46±9.26	95.52±5.81	**100.00±0.00**	97.42±7.44	**100.00±0.00**
OA(%)	71.99 ± 2.53	82.11 ± 2.81	73.37 ± 10.22	97.16±0.97	98.26±0.62	95.48±1.17	98.17±0.82	98.18±0.47	97.74±1.13	**99.48±0.22**
AA(%)	60.68 ± 3.08	83.54 ± 2.18	65.28 ± 12.86	95.14±1.34	96.78±1.18	95.49±1.35	98.51±0.74	96.26±0.67	95.70±2.07	**99.32±0.36**
$\kappa \times 100$	68.33 ± 2.97	80.12 ± 3.13	69.64 ± 12.37	96.83±1.08	98.07±0.69	94.97±1.30	97.96±0.91	97.98±0.52	97.48±1.26	**99.42±0.25**

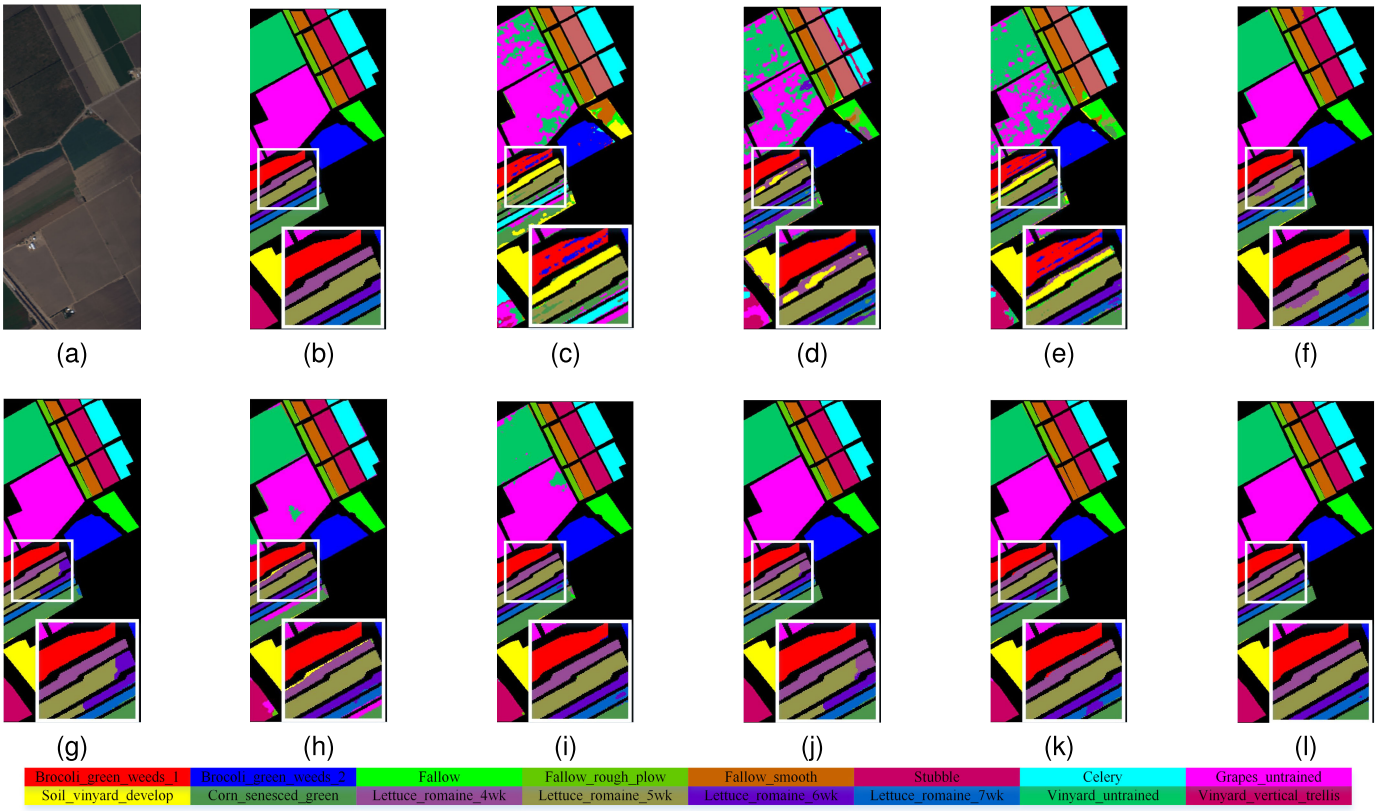


Fig. 6. False-color image, ground truth, and classification maps obtained by different methods on the Salinas dataset. (a) False-color image. (b) Ground Truth. (c) M3D-DCNN. (d) DFFN. (e) RSSAN. (f) GCN. (g) MDGCN. (h) CADGCN. (i) CEGCN. (j) DAGCN. (k) DNAGSI. (l) SDGCP.

with standard deviations provided to assess network stability.

1) Global Feature Extraction Subnetwork: It is evident that the proposed GFEsN exhibits significant advantages over the typical GCN by extracting global information from HSI, with an OA improvement of 3.73%, 1.14%, and 1.19% on the three datasets compared to the baseline. Due to the complex spatial structures in the Indian Pines dataset, GFEsN effectively utilizes high-order neighbor information, leading to significant performance enhancement on this dataset. In contrast, the Salinas and WHU-Hi-LongKou datasets, which are characterized by large homogeneous regions, already provide relatively sufficient spatial context information through low-order neighbor nodes, resulting in a smaller performance improvement. The above results demonstrate the superiority of GFEsN.

2) Prototype Learning Module: With the introduction of PLM, the classification accuracy on three datasets is improved. Moreover, the standard deviation of the results under ten experiments decreased, which indicates that the stability of the network is also enhanced. PLM precisely optimizes intraclass

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE WHU-HI-LONGKOU DATASET

Class	M3D-DCNN	DFFN	RSSAN	GCN	MDGCN	CADGCN	CEGCN	DAGCN	DNAGSI	SDGCP
1	96.94 ± 1.92	97.23 ± 0.52	98.85 ± 0.98	97.91±1.69	98.27±1.29	99.52±0.49	99.86±0.11	97.83±1.67	98.80±0.74	**99.88±0.13**
2	62.71 ± 7.21	78.12 ± 8.82	62.93 ± 32.31	93.31±5.05	99.05±1.66	91.51±10.54	94.90±5.25	96.91±4.33	92.94±7.27	**99.35±1.11**
3	0.00 ± 0.00	19.99 ± 30.32	30.37 ± 37.73	93.48±6.47	95.71±5.28	90.92±8.77	85.80±16.30	91.71±9.78	92.19±8.65	**97.66±2.29**
4	95.89 ± 0.97	97.24 ± 1.12	98.25 ± 0.81	97.78±1.08	98.00±0.90	98.77±0.39	99.45±0.25	98.09±1.08	98.49±1.60	**99.52±0.20**
5	12.15 ± 7.25	14.28 ± 13.21	30.23 ± 29.51	65.96±12.27	88.99±14.86	82.92±11.38	**89.90±10.34**	87.07±10.98	82.14±15.85	89.86±10.07
6	91.53 ± 4.16	87.04 ± 7.14	80.51 ± 33.10	96.00±3.42	97.41±3.12	96.56±3.61	98.83±2.18	96.43±3.93	96.42±2.57	**99.82±0.16**
7	**99.99 ± 0.01**	99.87 ± 0.14	99.83 ± 0.21	99.81±0.43	99.93±0.08	99.94±0.04	99.97±0.03	99.69±0.64	99.96±0.07	**99.99±0.01**
8	80.36 ± 4.80	80.04 ± 3.97	65.90 ± 33.09	77.12±12.53	76.92±13.51	86.07±16.03	86.84±17.07	78.99±14.45	79.12±10.41	**87.71±15.22**
9	19.43 ± 4.18	27.61 ± 16.44	28.40 ± 23.47	70.16±8.13	78.46±8.82	75.78±9.34	73.37±11.06	**78.47±6.55**	63.99±13.81	78.17±10.52
OA(%)	90.18 ± 0.80	91.52 ± 0.78	91.10 ± 5.23	96.05±0.63	97.24±0.76	97.39±0.83	97.96±1.00	96.94±0.63	96.70±0.78	**98.56±0.58**
AA(%)	62.11 ± 1.80	66.82 ± 4.53	66.14 ± 17.79	87.95±2.00	92.53±2.14	91.33±2.80	92.10±4.53	91.69±2.04	89.34±2.90	**94.66±2.10**
$\kappa \times 100$	86.91 ± 1.09	88.73 ± 1.07	88.02 ± 7.29	94.80±0.83	96.37±1.00	96.56±1.10	97.31±1.33	95.98±0.83	95.64±1.02	**98.11±0.77**

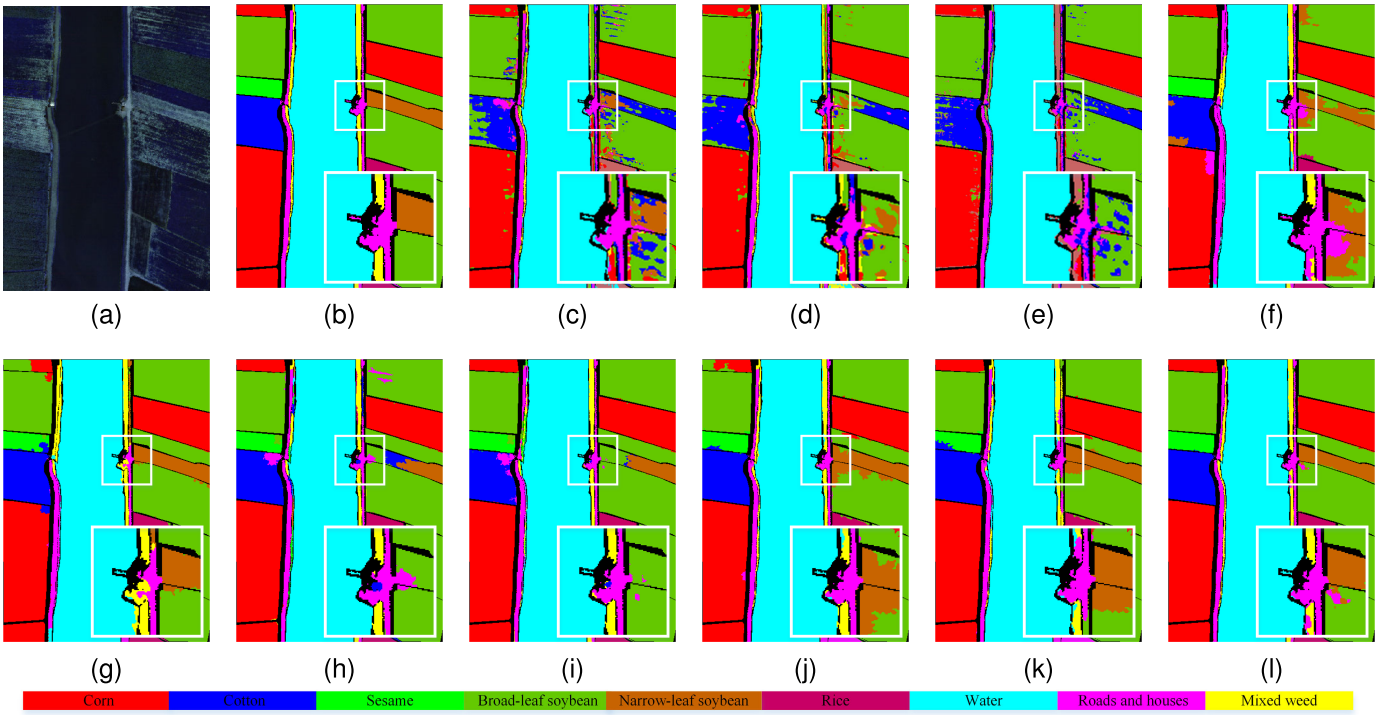


Fig. 7. False-color image, ground truth, and classification maps obtained by different methods on the WHU-Hi-LongKou dataset. (a) False-color image. (b) Ground Truth. (c) M3D-DCNN. (d) DFFN. (e) RSSAN. (f) GCN. (g) MDGCN. (h) CADGCN. (i) CEGCN. (j) DAGCN. (k) DNAGSI. (l) SDGCP.

TABLE VII
OA INDICES OF ABLATION EXPERIMENTS ON DIFFERENT DATASETS

	GCN+Softmax	GFEsN+Softmax	GFEsN+PLM	S^2AsN+GFEsN+PLM	S^2AsN+LFEsN+PLM	ALL
Indian Pines	85.08 ±2.60	88.81±2.75	89.47±2.12	89.64±2.09	73.89±4.08	**91.92±1.88**
Salinas	97.16±0.97	98.3±0.81	98.5±0.63	98.68±0.71	93.91±0.79	**99.48±0.22**
WHU-Hi-LongKou	96.05±0.63	97.24±1.02	97.49±0.61	97.83±0.77	97.37±0.82	**98.56±0.58**

and interclass distances between training samples, leading to a metric space where accurate classification can be achieved based on the distances between samples and prototypes of each class. This module exhibits greater robustness compared to the softmax classifier.

3) Spectral–Spatial Attention Subnetwork: Further enhancement in classification results is achieved by combining S^2AsN, GFEsN, and PLM, indicating the necessity of feature refinement for the original HSI. S^2AsN can adaptively select important spatial and spectral features while suppressing unnecessary features, which facilitates spatial and spectral feature representation and final classification.

4) Local Feature Extraction Subnetwork: In comparison to the baseline, LFEsN exhibits lower classification performance on the Indian Pines and Salinas datasets due to limited training samples. However, on the WHU-Hi-LongKou dataset, it surpassed the baseline, even outperforming some GCN-based methods listed in Table VI, further substantiating the effectiveness of S^2AsN and PLM. By introducing LFEsN to extract local detail information, which is complementary

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.

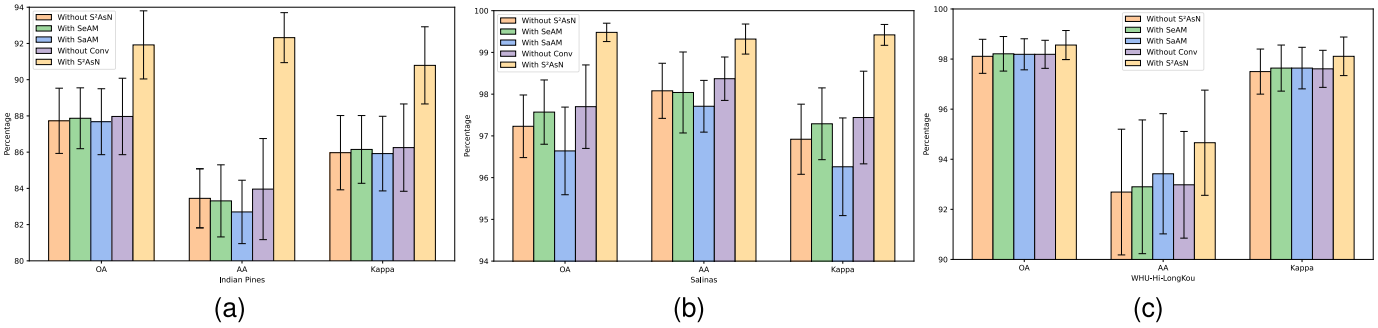


Fig. 8. Influence of different components in S^{2}AsN on three datasets. (a) Indian Pines. (b) Salinas. (c) WHU-Hi-LongKou.

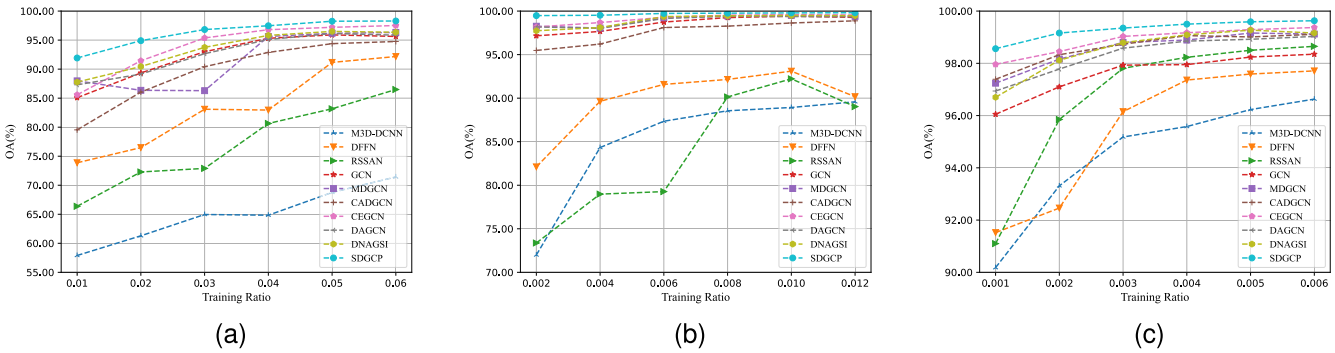


Fig. 9. Classification performance of each method with different training ratios on three datasets. (a) Indian Pines. (b) Salinas. (c) WHU-Hi-LongKou.

to the global information extracted by GFEsN, the proposed SDGCP achieves higher classification performance across all three datasets.

B. Discussion of S^2AsN

To further explore the contribution of the proposed S^{2}AsN to the overall model, we also conducted ablation analysis over its different components, namely, the spectral convolution, SeAM, and SaAM. The experimental results over three datasets are shown in Fig. 8. It can be observed that merely incorporating attention mechanisms did not significantly improve the network's performance, and in some cases, it even decreased. There are two main aspects to elucidate the reasons. First, from the perspective of network architecture, the output of S^{2}AsN needs to be encoded into a graph structure and then inputted into the SDGCM and LFEsN to extract global and local features. Since the original HSI contains noise and redundant information, directly adding an attention module inevitably processes these redundant information and noise, which limits the effectiveness in subsequent feature extraction. Second, by incorporating two layers of spectral convolution, a certain amount of learnable parameters can be added in S^{2}AsN. One direct advantage of this approach is that while performing dimensionality reduction and denoising on the original HSI, it reduces the number of learnable parameters in the attention module, LFEsN, and GFEsN. The additional parameters introduced in the S^{2}AsN not only reduce the parameter count in the subsequent feature extraction network but also make the refined features more conducive to subsequent feature extraction.

C. Influence of Label Ratio

Experiments on three datasets are conducted to validate that the proposed SDGCP outperforms other comparative algorithms in terms of classification performance at different training ratios. Specifically, 1%–6% labeled samples are randomly chosen as training data for the Indian Pines dataset, 0.2%–0.12% labeled samples are randomly chosen as training data for the Salinas dataset, and 0.1%–0.6% labeled samples are randomly chosen as training data for the WHU-Hi-LongKou dataset. As shown in Fig. 9, the proposed SDGCP achieves the best classification performance among the three datasets. On the Indian Pines and WHU-Hi-LongKou datasets, SDGCP maintains a notable gap in OA compared to other methods. However, on the Salinas dataset, the gap decreases as the number of training samples increases, which highlights the strong superiority of the proposed SDGCP under limited labeled samples. From Fig. 9, it is evident that methods based on GCN outperform those based on CNN in terms of classification accuracy when the training sample ratio is small. However, as the training ratio increased, the improvement in OA is not as rapid as that of CNN-based methods. DFFN outperforms GCN in terms of classification performance on the WHU-Hi-LongKou dataset when the training ratio exceeds 0.4%, which indicates that CNN has a greater advantage when there are sufficient training samples. By integrating the advantages of CNN and GCN, SDGCP can obtain satisfactory classification accuracy under different training ratios.

D. Visualization and Analysis of Feature Fusion

The proposed SDGCP significantly enhances the classification performance by integrating GCN with CNN. To explore

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.

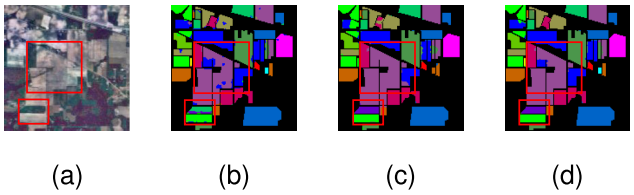


Fig. 10. Visualization of feature fusion on Indian Pines dataset. (a) False–color image. (b) Visualization of all areas by LFEsN. (c) Visualization of all areas by GFEsN. (d) Visualization of all areas by SDGCP.

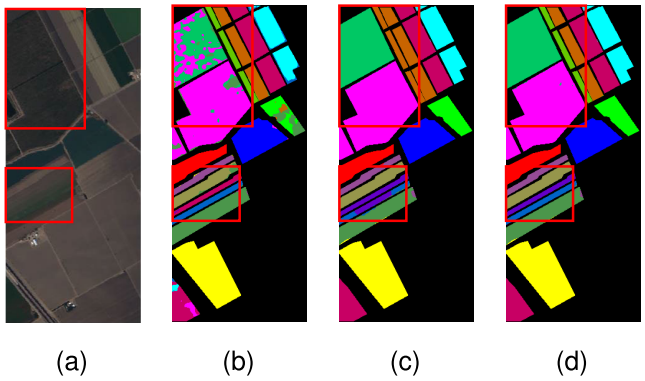


Fig. 11. Visualization of feature fusion on Salinas dataset. (a) False-color image. (b) Visualization of all areas by LFEsN. (c) Visualization of all areas by GFEsN. (d) Visualization of all areas by SDGCP.

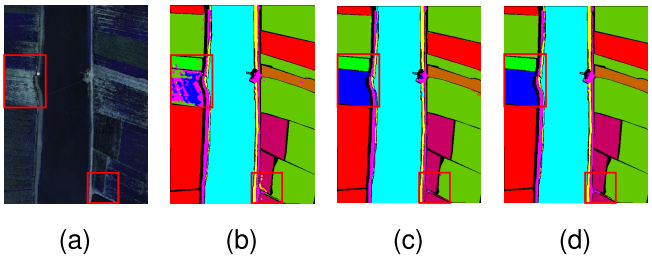


Fig. 12. Visualization of feature fusion on WHU-Hi-LongKou dataset. (a) False-color image. (b) Visualization of all areas by LFEsN. (c) Visualization of all areas by GFEsN. (d) Visualization of all areas by SDGCP.

the effectiveness of the local features extracted by LFEsN and the global features extracted by GFEsN within the network, we visualize the classification results of the two branches in Figs. 10–12, and regions showing notable differences are marked by red boxes. The LFEsN based on CNN is capable of correctly classifying at region boundaries but exhibits a "salt and pepper" feature in large homogeneous areas. The classification map generated by GFEsN based on GCN exhibits smoother boundaries, but at the expense of misclassifications in smaller regions. After fusing the features extracted by both branches, it is evident that the proposed SDGCP effectively combines the strengths of CNN and GCN, resulting in a more precise classification map.

E. Feature Visualization With Different Loss Functions

To intuitively validate the effectiveness of the proposed PLM, different loss functions are employed to visualize features of the Indian Pines dataset, including softmax-based cross-entropy loss, DCE loss, and PLM loss. The visualized

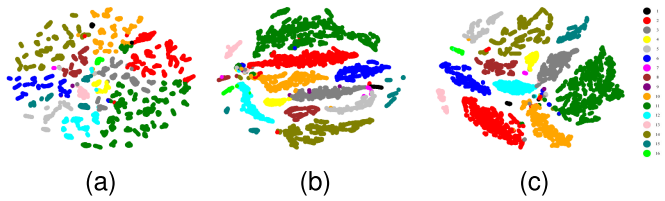


Fig. 13. Feature visualization with different loss functions on the Indian Pines dataset. (a) Softmax. (b) DCE. (c) PLM.

TABLE VIII
IMPACT OF NETWORK DEPTH ON DIFFERENT DATASETS

Layer	2	4	8	16	32	64	128
Indian Pines	90.7	91.01	91.08	91.46	91.55	**91.92**	91.52
Salinsa	99.12	99.3	99.31	99.39	99.44	**99.48**	99.44
WHU-Hi-LongKou	98.36	98.39	98.36	98.42	98.43	**98.56**	98.48

features of the testing samples encompass both the features generated by the network before the softmax function and the embedded features generated by the PLM. T-SNE [57] is employed as the visualization technique and the visualization results are displayed in Fig. 13. It can be observed that the features generated without using PLM exhibit more considerable interclass distances compared to intraclass distances, resulting in difficulties for subsequent classification. Features generated using the DCE loss are more concentrated. Moreover, when augmented with L_{foc} and L_{sep} regularization terms, the extracted features exhibit both intraclass compactness and interclass separability. For example, as shown in Fig. 13(b), there is an overlap between classes 1, 3, 4, 5, and 7, while classes 1, 4, 5, and 7 in Fig. 13(c) are separated from class 3. Consequently, such a designed loss function enables the proposed SDGCP to achieve more accurate classification at the boundaries of different land cover classes.

F. Hyperparameter Analyses

1) Influence of Network Depth: The depth of the network directly relates to the higher order neighbor information that can be utilized. To explore the impact of network depth, SDGCP with different network depths are tested on three datasets, with each experiment repeated ten times. As shown in Table VIII, it can be observed that the classification performance is poor when the network is too deep or too shallow. As the network depth increases, the OA of SDGCP gradually increases and reaches its maximum when L is 64, which is attributed to the ability of deep networks to capture features from high-order neighboring nodes. Subsequently, the OA value begins to decrease. This is because the network depth exceeds a critical point, causing the information from neighboring nodes used by central nodes to cover the entire graph, leading to a gradual convergence of features between nodes and eventually resulting in misclassification. It is noteworthy that the network depth adopted by SDGCP is set to 64.

2) Influence of Segmentation Scale: The scale of superpixel segmentation determines the number of graph nodes, thus affecting both computational efficiency and classification

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.

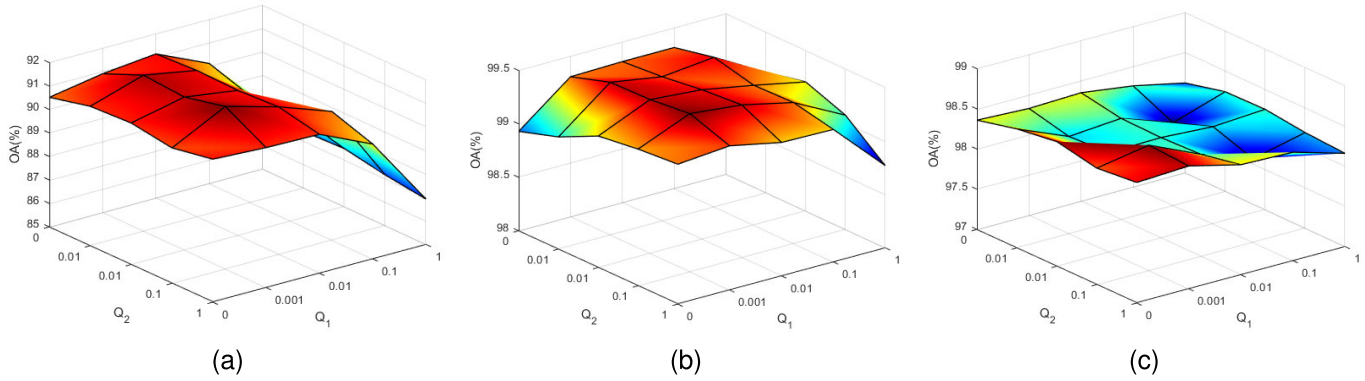


Fig. 14. Parameter sensitivity analysis of the regularizer losses on three datasets. (a) Indian Pines. (b) Salinas. (c) WHU-Hi-LongKou.

TABLE IX
CLASSIFICATION ACCURACY OF SDGCP WITH DIFFERENT SEGMENTATION SCALES ON EACH DATASET

Scale	50	100	150	200	250	300
Indian Pines	**91.92**	91.36	91.44	90.81	90.76	90.38
Salinsa	98.91	99.35	99.41	99.46	99.42	**99.48**
WHU-Hi-LongKou	98.46	98.48	98.4	98.47	98.46	**98.56**

TABLE X
INFLUENCE OF α ON THREE DATASETS

α	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
Indian Pines	**91.92**	91.08	90.90	89.89	89.09	86.84	84.84	81.53	77.59
Salinsa	**99.48**	99.45	99.34	99.28	99.12	98.97	98.15	97.02	95.12
WHU-Hi-LongKou	**98.56**	98.52	98.38	98.51	98.39	98.24	98.12	97.93	97.46

results. To investigate the influence of different superpixel segmentation scales, λ is set to 50, 100, 150, 200, 250, and 300 on three datasets, and the test results on the proposed SDGCP are shown in Table IX. It can be observed that OA exhibits an overall decreasing trend with the increase in segmentation scale on the Indian Pines dataset. This is attributed to the presence of small-sized land objects in the dataset, where larger superpixel blocks tend to encompass multiple classes of objects. In contrast to Indian Pines, for the Salinas dataset, larger segmentation scales result in better classification performance with SDGCP. This is because the Salinas dataset contains large homogeneous regions. The impact of segmentation scale on the WHU-Hi-LongKou dataset is relatively minimal. Taking into consideration both the number of graph nodes and classification performance, we select segmentation scales of 50, 300, and 300 for the three datasets in our experiments.

3) Influence of α: The hyperparameter α controls the proportion of features extracted by LFEsN in the fusion process. Table X exhibits the OA metrics on three datasets with α ranging from $\{0.1, 0.2, \ldots, 0.9\}$. It is evident that as α increases, the overall OA of SDGCP gradually decreases, particularly on the Indian Pines dataset. This demonstrates the superiority of GCN under the constraints of limited labeled training samples. In this article, the fusion ratio α is set to 0.1.

TABLE XI
INFLUENCE OF ω ON THREE DATASETS

ω	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
Indian Pines	**91.92**	91.55	90.91	90.02	89.07	87.68	86.02	84.03	83.48
Salinas	**99.48**	99.24	98.79	98.43	97.88	97.56	97.05	96.81	96.38
WHU-Hi-LongKou	98.56	98.52	**98.59**	98.58	98.54	98.43	98.39	98.24	98.17

4) Influence of ω: ω is a key factor in addressing oversmoothing issues, which determines how much of the initial graph's feature information is retained during each feature propagation. Table XI illustrates the impact of ω on the proposed SDGCP. On the Indian Pines and Salinas datasets, there is an overall decreasing trend in the OA metric as ω increases. However, on the WHU-Hi-LongKou dataset, the OA reaches its maximum value at ω equal to 0.3. A higher value of ω implies less interaction of information between different nodes in the feature propagation module. To enable SDGCP to capture more global spatial context information, we set ω to 0.1 on all three datasets.

5) Influence of Q_1 and Q_2: In the PLM, two critical hyperparameters, Q_1 and Q_2, determine the importance of interclass distance and intraclass distance, and experiments are conducted to analyze their impact on three different datasets. Keeping other parameters constant, the hyperparameters Q_1 and Q_2 are set to elements from the set {0.0001, 0.001, 0.01, 0.1, 1}, and the OA is shown in Fig. 14. It can be observed that the hyperparameter Q_1 plays a dominant role in the classification results on the Indian Pines and Salinas datasets, where the intraclass loss contributes more significantly, leading to poorer classification results when Q_1 is increased. However, on the WHU-Hi-LongKou dataset, the classification results fluctuate with changes in both Q_1 and Q_2, indicating that the model is less sensitive to Q_1 and Q_2 when dealing with this dataset. Taking all factors into consideration, the parameter combination (Q_1, Q_2) is set to (0.001, 0.1) in our proposed SDGCP method and applied to all datasets. This choice demonstrates good performance across various datasets.

G. Comparison of Running Time

The training and testing times of different comparative algorithms over the three benchmarks are elaborated in detail in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.

TABLE XII
RUNNING TIME (S) OF DIFFERENT METHODS ON THREE DATASETS

Dataset	Time	M3D-DCNN	DFFN	RSSAN	GCN	MDGCN	CADGCN	CEGCN	DAGCN	DNAGSI	SDGCP
Indian Pines	Train time(s)	8.26	15.38	11.3	**7.28**	115.42	17.43	15.97	18.97	344.62	20.76
	Test time(s)	1.58	4.42	2.89	**0.65**	1.67	1.02	0.67	1.13	1.7	0.82
Salinas	Train time(s)	**10.83**	18.7	12.87	12.22	398.71	56.14	25.7	24.44	277.48	33.55
	Test time(s)	5.5	18.06	10.89	**2.87**	6.56	5.58	5.4	5.52	3.59	3.9
WHU-Hi-LongKou	Train time(s)	29.49	35.99	**25.12**	31.69	1864.16	75.32	55.25	53.78	595.92	51.97
	Test time(s)	13.29	42.89	21.48	**5.44**	12.34	9.85	10.31	10.24	7.42	8.12

Table XII. CNN-based algorithms generally spend less training time since their input is a small number of training patches. By adopting a simple network structure, GCN spends the least testing time over all the other considered algorithms in all three datasets. For the proposed SDGCP, despite adopting a deeper structure, it exhibits training and testing times comparable to other shallow network methods. This is because the proposed SDGCM does not introduce additional learnable parameters during the process of deep feature propagation.

V. CONCLUSION

In this article, a framework called SDGCP, which includes an S^2AsN, a GFEsN, an LFEsN, and a PLM, is proposed for HSI classification. In the proposed SDGCP, S^2AsN is first constructed to perform dimensionality reduction on the original HSI and then adaptively select spectral–spatial features. Through the processing of the S^2AsN, the features are refined and subsequently fed into two branches of the network for feature extraction. One branch named LFEsN, which consists of spectral–spatial convolutional layers, is specifically tailored for capturing local information from the HSI. The other branch called GFEsN is an SDGCM composed of feature extraction and feature propagation, which is utilized to capture high-order neighbor node information in the superpixel graph, significantly solving the problem of oversmoothing and reducing the number of learnable parameters in deep GCN. After fusing long-range spatial context information with local information, PLM is introduced to enhance feature discriminability under limited training samples, which contributes to both intraclass compactness and interclass separability. Finally, extensive experiments on three datasets demonstrate that the proposed SDGCP not only surpasses other SOTA CNN and GNN methods in terms of classification performance but also exhibits superior model stability.

The proposed SDGCP employs an additive operation to fuse the global features extracted by GCN and the local features extracted by CNN. In the future, we will try to adaptively fuse these two types of features through the attention mechanism.

REFERENCES

[1] Y. Lanthier, A. Bannari, D. Haboudane, J. R. Miller, and N. Tremblay, "Hyperspectral data segmentation and classification in precision agriculture: A multi-scale analysis," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, vol. 2, 2008, pp. 585–588.

[2] F. van der Meer, "Analysis of spectral absorption features in hyperspectral imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 5, no. 1, pp. 55–68, Feb. 2004.

[3] T. Schmid, M. Koch, and J. Gumuzzio, "Multisensor approach to determine changes of wetland characteristics in semiarid environments (central Spain)," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 11, pp. 2516–2525, Nov. 2005.

[4] T. J. Malthus and P. J. Mumby, "Remote sensing of the coastal zone: An overview and priorities for future research," *Int. J. Remote Sens.*, vol. 24, no. 13, pp. 2805–2815, 2003.

[5] D. Manolakis, D. Marden, and G. A. Shaw, "Hyperspectral image processing for automatic target detection applications," *Lincoln Lab. J.*, vol. 14, no. 1, pp. 79–116, 2003.

[6] S. Mei, J. Ji, J. Hou, X. Li, and Q. Du, "Learning sensor-specific spatial–spectral features of hyperspectral images via convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4520–4533, Aug. 2017.

[7] S. Mei, J. Ji, Y. Geng, Z. Zhang, X. Li, and Q. Du, "Unsupervised spatial–spectral feature learning by 3D convolutional autoencoder for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6808–6820, Sep. 2019.

[8] F. Xu, G. Zhang, C. Song, H. Wang, and S. Mei, "Multiscale and cross-level attention learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501615.

[9] S. Mei, X. Li, X. Liu, H. Cai, and Q. Du, "Hyperspectral image classification using attention-based bidirectional long short-term memory network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.

[10] S. Mei, X. Chen, Y. Zhang, J. Li, and A. Plaza, "Accelerating convolutional neural network-based hyperspectral image classification by step activation quantization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.

[11] J. Wang, W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Hyperspectral and SAR image classification via multiscale interactive fusion network," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–15, vol. 34, no. 12, pp. 10823–10837, 2023.

[12] W. Li, J. Wang, Y. Gao, M. Zhang, R. Tao, and B. Zhang, "Graph-feature-enhanced selective assignment network for hyperspectral and multispectral data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526914.

[13] J. Wang, M. Zhang, W. Li, and R. Tao, "A multistage information complementary fusion network based on flexible-mixup for HSI-X image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, pp. 1–13, 2024.

[14] Y. Zhong and L. Zhang, "An adaptive artificial immune network for supervised classification of multi-/hyperspectral remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 3, pp. 894–909, Mar. 2012.

[15] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Semisupervised hyperspectral image segmentation using multinomial logistic regression with active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4085–4098, Nov. 2010.

[16] E. Blanzieri and F. Melgani, "Nearest neighbor classification of remote sensing images with the maximal margin principle," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 6, pp. 1804–1811, Jun. 2008.

[17] D. A. Landgrebe, *Signal Theory Methods in Multispectral Remote Sensing*, vol. 24. Hoboken, NJ, USA: Wiley, 2003.

[18] P. K. Goel, S. O. Prasher, R. M. Patel, J. A. Landry, R. B. Bonnell, and A. A. Viau, "Classification of hyperspectral data by decision trees and artificial neural networks to identify weed stress and nitrogen status of corn," *Comput. Electron. Agricult.*, vol. 39, no. 2, pp. 67–93, May 2003.

[19] S. Prasad and L. M. Bruce, "Limitations of principal components analysis for hyperspectral target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 4, pp. 625–629, Oct. 2008.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.

[20] A. Villa, J. A. Benediktsson, J. Chanussot, and C. Jutten, "Hyperspectral image classification with independent component discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 4865–4876, Dec. 2011.

[21] T. V. Bandos, L. Bruzzone, and G. Camps-Valls, "Classification of hyperspectral images with regularized linear discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 3, pp. 862–873, Mar. 2009.

[22] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[23] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[24] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, 2015.

[25] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[26] S. Zhang, Y. Li, and S. Mei, "Exploring uni-modal feature learning on entities and relations for remote sensing cross-modal text-image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5626317.

[27] J. Lian, S. Mei, S. Zhang, and M. Ma, "Benchmarking adversarial patch against aerial detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5634616.

[28] J. Lian, X. Wang, Y. Su, M. Ma, and S. Mei, "CBA: Contextual background attack against optical aerial detection in the physical world," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5606616.

[29] J. Wang, F. Li, Y. An, X. Zhang, and H. Sun, "Towards robust LiDAR-camera fusion in BEV space via mutual deformable attention and temporal aggregation," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Feb. 16, 2024, doi: 10.1109/TCSVT.2024.3366664.

[30] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[31] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[32] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.

[33] M. He, B. Li, and H. Chen, "Multi-scale 3D deep convolutional neural network for hyperspectral image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3904–3908.

[34] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2017.

[35] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.

[36] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.

[37] J. Wang, W. Li, M. Zhang, and J. Chanussot, "Large kernel sparse ConvNet weighted by multi-frequency attention for remote sensing scene understanding," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5626112.

[38] J. Wang, W. Li, M. Zhang, R. Tao, and J. Chanussot, "Remote-sensing scene classification via multistage self-guided separation network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5615312.

[39] M. Zhang, W. Li, X. Zhao, H. Liu, R. Tao, and Q. Du, "Morphological transformation and spatial-logical aggregation for tree species classification using hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501212.

[40] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 1912–1925, 2024.

[41] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 6, pp. 2817–2830, 2023.

[42] Y. Zhang, M. Zhang, W. Li, S. Wang, and R. Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501312.

[43] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1498–1512, 2023.

[44] J. Wang, W. Li, Y. Wang, R. Tao, and Q. Du, "Representation-enhanced status replay network for multisource remote-sensing image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, pp. 1–13, 2023.

[45] A. Qin, Z. Shang, J. Tian, Y. Wang, T. Zhang, and Y. Y. Tang, "Spectral–spatial graph convolutional networks for semisupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 241–245, Feb. 2018.

[46] L. Mou, X. Lu, X. Li, and X. X. Zhu, "Nonlocal graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8246–8257, Dec. 2020.

[47] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2019.

[48] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[49] Q. Li, Z. Han, and X.-M. Wu, "Deeper insights into graph convolutional networks for semi-supervised learning," in *Proc. AAAI Conf. Artif. Intell.*, 2018, vol. 32, no. 1, pp. 3538–3545.

[50] B. Xi et al., "Semisupervised cross-scale graph prototypical network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 9337–9351, 2023.

[51] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel- and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.

[52] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.

[53] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2016, *arXiv:1609.02907*.

[54] S. Wan, C. Gong, P. Zhong, S. Pan, G. Li, and J. Yang, "Hyperspectral image classification with context-aware dynamic graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 597–612, Jan. 2021.

[55] J. Bai, B. Ding, Z. Xiao, L. Jiao, H. Chen, and A. C. Regan, "Hyperspectral image classification based on deep attention graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504316.

[56] B. Yang, F. Cao, and H. Ye, "A novel method for hyperspectral image classification: Deep network with adaptive graph structure integration," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5523512.

[57] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 1–27, 2008.



Yingjie Lu (Graduate Student Member, IEEE) received the B.Eng. degree in electronics and information engineering from Northwestern Polytechnical University, Xi'an, China, in 2022, where he is currently pursuing the M.Eng. degree in information and communication engineering with the School of Electronics and Information.

His main research interests include remote sensing and deep learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.



Shaohui Mei (Senior Member, IEEE) received the B.S. degree in electronics and information engineering and the Ph.D. degree in signal and information processing from Northwestern Polytechnical University, Xi'an, China, in 2005 and 2011, respectively.

He was a Visiting Student with the University of Sydney, Camperdown, NSW, Australia, from October 2007 to October 2008. He is currently a Professor with the School of Electronics and Information, Northwestern Polytechnical University. His research interests include hyperspectral remote sensing image processing and applications, intelligent signal and information acquisition and processing, video processing, and pattern recognition.

Dr. Mei received the First Prize of Natural Science Award of Shaanxi Province in 2022, the Excellent Doctoral Dissertation Award of Shaanxi Province in 2014, the Best Paper Award of the IEEE International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS) in 2017, and the Best Reviewer of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS) in 2019 and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) in 2022. He is serving as an Associate Editor for IEEE TGRS and IEEE JSTARS, a Guest Editor for *Remote Sensing*, and a reviewer for more than 30 international famous academic journals.



Fulin Xu received the B.S. degree from Northwestern Polytechnical University, Xi'an, China, in 2021, where he is currently pursuing the master's degree in signal and information processing.

His research interests include hyperspectral remote sensing image processing and deep learning.



Mingyang Ma (Graduate Student Member, IEEE) received the B.Eng. degree in communication engineering and the Ph.D. degree in communication and information systems from Northwestern Polytechnical University, Xi'an, China, in 2015 and 2021, respectively.

His main research interests include video summarization and image processing.



Xiaofei Wang (Graduate Student Member, IEEE) received the B.Eng. degree in electronics and information engineering from Northwestern Polytechnical University, Xi'an, China, in 2022, where she is currently pursuing the M.Eng. degree in information and communication engineering with the School of Electronics and Information.

Her main research interests include remote sensing and image processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:30:33 UTC from IEEE Xplore. Restrictions apply.